Feasibility Assessment of Balanced-Model EV Charging-Flexibility Schedules in Unbalanced Three-Phase Distribution Networks

A balanced-model flexibility offer is a plan, not yet a dispatchable product.

Validate the fixed vehicle–time allocation on the detailed feeder, then return the largest safe fraction α^* before dispatch.

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1 Why balanced offers can fail

The implementation gap

EV flexibility is often planned with a balanced or aggregate network model. After the original feeder phases, EV locations, and load imbalance are restored, the same vehicle-level V1G schedule can exceed voltage, VUF, transformer, or feeder-head limits.

The question

How much of a fixed vehicle-level offer can be dispatched safely?

The goal is not to redesign which EV charges when. It is to quantify the network headroom left in an already computed schedule.

The output

For every case, the procedure returns two operational quantities:

- **the largest verified feasible factor $\alpha^* \in [0, 1]$;**
- **the first network constraint that blocks a larger offer.**

Evidence below covers 46 evaluated scenarios on two benchmark feeders; it is case-study evidence, not a universal guarantee.

2 From a planned schedule to a safe dispatch

The schedule-preserving gate keeps the original EV–time pattern and changes only its implementation level. Vehicle feasibility is inherited from the two endpoints under the continuous V1G model; network feasibility is re-tested in the detailed unbalanced feeder.

1. **Baseline** availability, energy demand, charging limits
2. **Target** balanced-model vehicle-level V1G schedule
3. **Scale** $u(\alpha) = u^0 + \alpha(u^B - u^0)$
4. **Check** 96-slot nonlinear three-phase power flow
5. **Return** largest passing α^* + first limit

$$u_{i,t}(\alpha) = u^0_{i,t} + \alpha\left(u^B_{i,t} - u^0_{i,t}\right), \quad 0 \le \alpha \le 1$$

Both endpoints satisfy availability, power bounds, cumulative-energy, and departure-target constraints; intermediate continuous V1G schedules preserve the same vehicle–time allocation.

Network gate at every 15-min slot

- $0.95 \le V \le 1.06$ **p.u.**
- **VUF** $\le 2\%$
- **transformer loading** ≤ 1.0 **p.u.**
- **feeder-head loading** ≤ 1.0 **p.u.**

How to read α^*: **If $\alpha^* = 1$, the target is fully dispatchable. If $\alpha^* = 0.54$, dispatch 54% of the original adjustment while keeping the same EV–time pattern; the remaining 46% is withheld at the first feeder limit.**

3 Study design

Restore the feeder detail



IEEE13



LV906

OpenDSS validation restores phase assignments, feeder imbalance, and EV connection points before the three-phase check.

Case matrix

22 + 24 = 46

IEEE13 + LV906 evaluated cases.

96×501

quarter-hour slots × candidate α values per case.

Four call times: 00:00, 02:00, 06:00, and 19:00; six EV realizations per feeder–call-time pair.

EV inputs are sampled from 3,568 NHTS vehicle-day records.

Feeder comparison

	cases	mean α^*	median	full
IEEE13	22	0.834	0.999	11/22
LV906	24	0.540	0.446	3/24
Overall	46	0.681	0.719	14/46

A full result means $\alpha^* = 1$. The study deliberately compares a small meshed benchmark with a larger low-voltage feeder where phase imbalance has more room to accumulate.

The validation is schedule-specific: each candidate must pass every active constraint at every time slot before it is called dispatchable.

4 What changes at dispatch?

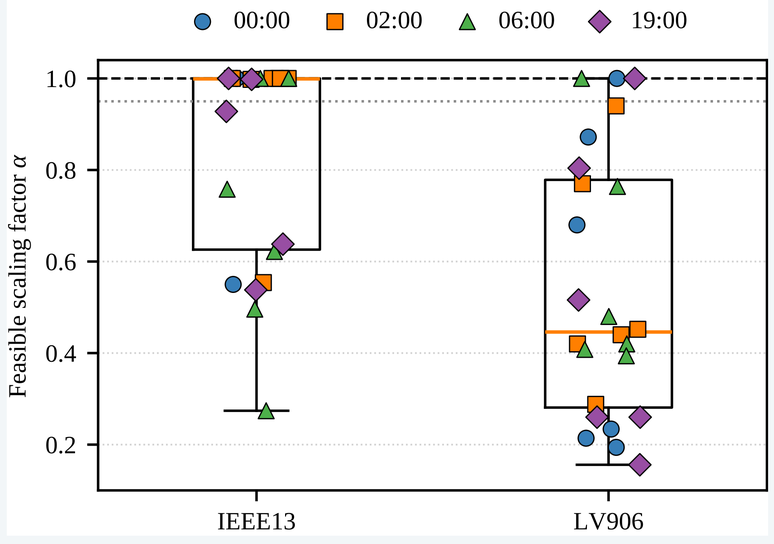
The balanced target is often only partially deliverable. Derating is not a change in the EV pattern: it is a network-feasibility correction applied before the offer reaches dispatch.

32/46
target schedules require derating

14/46
fully implementable at $\alpha^* = 1$

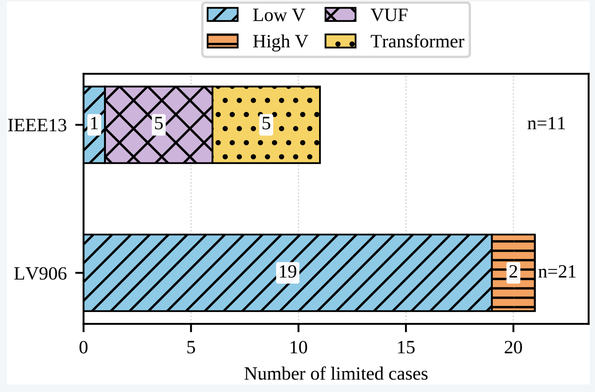
27/32
limited cases first blocked by voltage or VUF

Verified scaling factors



IEEE13 retains more headroom (mean $\alpha^* = 0.834$) than LV906 (mean $\alpha^* = 0.540$).

First limiting constraint



Voltage magnitude or VUF is the first limit in 27 of the 32 derated cases.

Why it matters

Similar scheduled offers can yield different verified offers because phase imbalance and network headroom differ.

How to read the offer

$\alpha^* = 1.00$	**100% of the planned adjustment is dispatchable.**
$\alpha^* = 0.54$	**54% is dispatchable; 46% is withheld at the first limit.**

LV906: 21/24 cases are limited; 19 by low voltage and 2 by high voltage.

Operational message **Validate → derate → dispatch. The procedure preserves the original vehicle–time allocation while making feeder headroom explicit; rerun it when feeder or EV states change.**

Evidence from 46 evaluated scenarios; the first limiting constraint is scenario-specific.

github.com/Kato814/EV-FLEX-BIAS